

[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

Dec-25-0491

MCA-6101 (Programming in C)

MCA 1st (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all, selecting one question each from Section A, B, C and D. Question no. 9 is compulsory.

SECTION - A

1. (a) Explain the general structure of a C program with the help of a neat diagram. Describe the purpose of each section briefly. (6)
(b) Write a program in C to find Fibonacci Series of n number. (6)
2. (a) Explain the different data types available in C. Classify them into basic, derived, enumeration, and user-defined types. (6)
(b) Write a C program to list first n prime numbers. (6)

SECTION - B

3. (a) What are the scope rules of functions in C? Explain with an example. (6)
(b) Define array. Write a C program to find the sum and average of n elements in an array. (6)
4. (a) Explain the use of malloc() and free() functions to allocate and de allocate memory dynamically. (6)

2

MCA-6101

- (b) Draw a flowchart and write a program to perform linear search in an array of n elements. (6)

SECTION - C

5. (a) Write a program that reads a string and counts number of vowels, words present in string. (6)
(b) Write a program to check whether entered string is palindrome or not. (6)
6. Define a structure in C to store details of an employee such as Employee ID, Name, Age, Department, and Salary. Write a C program to:
 - i. Declare an array of 5 employees using the structure.
 - ii. Accept details of all employees from the user.
 - iii. Display the details of employees who earn a salary greater than 30,000. (3×4=12)

SECTION - D

7. (a) Write a program in C to enter strings of data from keyboard and writes them onto a disk file. (6)
(b) Discuss various input and output operations of files. (6)
8. (a) What is the need for creating user-defined header files in C? Explain the steps involved in creating and using a header file with an example. (6)
(b) Define macros in C. Differentiate between object-like macros and function-like macros with suitable examples. (6)

[P.T.O.]

SECTION - E (Compulsory)

9. (i) What are C instructions?
- (ii) Discuss unformatted function.
- (iii) What is array of pointers?
- (iv) What is the purpose of fprintf() function?
- (v) Discuss bitwise operator.
- (vi) Find largest number between three numbers using ternary operator. (6×2=12)